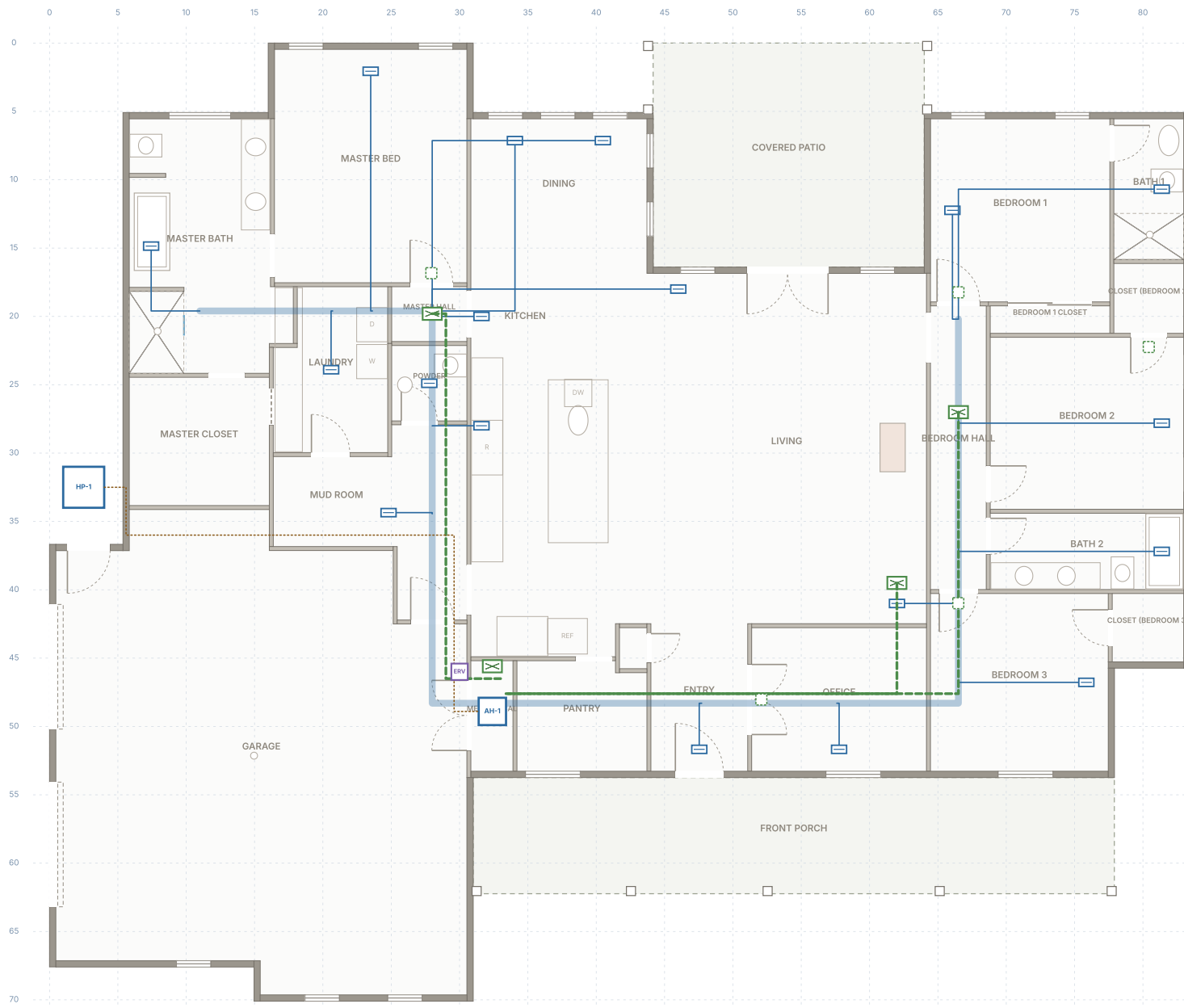




64 STONE SHEEP CIRCLE

From the owner's plans, Larson House 6/11/26

- ## HVAC
- Room loads at Casper, WY (assumed) design conditions, one ducted heat pump with backup heat, ducts in the attic.
- Supplies blue on the trunks; returns green dashed.
 - AH-1 in the mechanical closet, HP-1 outside, ERV-1 for fresh air.
 - 3.5 tons · heat 39 kBtu/h · cool 21 kBtu/h

Top of sheet = rear (patio). Bottom = front (porch).
Grid: 5 ft, feet from the garage door wall (X) and from the master bedroom's rear wall (Y).

M-1 HVAC · 64 Stone Sheep Circle

Room loads at Casper, WY (assumed) design conditions, one ducted heat pump with backup heat, ducts in the attic. App version v1. From the owner's plans (Larson House, 6/11/26).

Notes

1. Design conditions: Casper, WY (assumed) (ASHRAE 2009, Casper Natrona Co Intl AP, 99.6 % heating / 1 % cooling). Heating -10.3°F, cooling 89.6°F; inside 70°F / 75°F. Location to confirm: these change with it.
2. Envelope: walls U-0.045 (R-20 + R-5 ci), ceiling R-49 (vault R-38), windows U-0.3 SHGC 0.4, slab edge R-10, 3.0 ACH50 (IECC 2021 zone 6).
3. One ducted cold-climate heat pump; the air handler fits the 3'x8' mechanical closet (about 22"x22" footprint, return plenum below or beside). Trunks run in the attic over the flat ceilings: east along the front band and up the bedroom hall, west over the mud room and master hall.
4. The kitchen and living vault (4:12) has no flat attic over it: its supplies are high-sidewall grilles fed from the flat-ceiling attics beside it, or soffit runs.
5. Ducts in the attic: sealed and tested ($\leq$ 4 CFM25 per 100 sf), R-8 insulation, buried under the attic insulation where possible.
6. Every bedroom gets a return path (transfer grille or jump duct). Bath fans and the range hood exhaust outside (E-1); a range hood over 400 CFM needs makeup air.
7. The wood fireplace insert needs its own outside combustion-air kit and a listed chimney.

Open items

- Location: the design temperatures, and with them every number on this sheet, use Casper, WY until the site is confirmed.
- Fuel: heat pump with electric backup is assumed. A gas furnace with AC is the other common choice; it needs a gas line, a flue and combustion air in the closet.
- Mechanical closet opens from the garage: confirm the AHJ allows the air handler there, and air-seal it (gasketed door, sealed plenum, no return openings).
- Insulation values and air-tightness target for the house.
- Outdoor unit location (drawn by the master closet's outside wall, out of sight from the front).

Room loads and airflow

Room	Floor sf	Ext. wall sf	Glass sf	Heat Btu/h	Cool Btu/h	CFM	Registers	Duct
Master Bed	240	102	25	2,724	1,768	99	1	7"
Master Bath	192	247	14	3,389	1,005	103	1	7"
Master Closet	96	84	0	1,335	291	40	0	—
Master Hall	22	0	0	105	36	3	0	—
Laundry	101	0	0	478	565	32	1	6"

Room	Floor sf	Ext. wall sf	Glass sf	Heat Btu/h	Cool Btu/h	CFM	Registers	Duct
Powder	30	0	0	145	50	4	1	6"
Mud Room	133	0	0	1,207	380	37	1	6"
Dining	136	149	62	3,707	3,089	172	2	7"
Kitchen	370	0	0	2,203	2,412	135	2	6"
Pantry	77	70	16	1,414	715	43	0	—
Mechanical	25	28	0	484	106	15	0	—
Living	524	111	25	5,781	3,194	178	2	7"
Entry	76	40	0	1,199	275	36	1	6"
Coat Closet	6	0	0	29	10	1	0	—
Office	134	99	16	1,937	1,600	89	1	7"
Bedroom Hall	86	0	0	408	141	12	0	—
Bedroom 1	176	214	25	3,360	1,776	102	1	7"
Bedroom 1 Closet	18	0	0	83	29	3	0	—
Bath 1	53	139	0	1,419	227	43	1	6"
Closet (Bedroom 2)	26	46	0	512	90	16	0	—
Bedroom 2	178	88	25	2,314	1,652	92	1	7"
Bath 2	79	50	0	799	181	24	1	6"
Bedroom 3	169	218	16	3,104	1,478	94	1	7"
Closet (Bedroom 3)	26	92	0	901	137	27	0	—

System

Item	Size	Basis
Heating load	39,035 Btu/h	70°F inside, -10.3°F outside
Cooling load	21,207 sensible + 1,697 latent Btu/h	75°F inside, 89.6°F outside
Heat pump	3.5 tons, 1,400 CFM	sized to cooling (Manual S); cold-climate model
Backup heat	5 kW	covers what the heat pump can't at the heating design temperature
Supply registers	18	4×10 to 6×12 ceiling, 6-8" R-8 flex
Ventilation	133 CFM ERV	ASHRAE 62.2 continuous

Bill of materials

Rough-order quantities from the layout, for pricing. The HVAC contractor takes off the final list.

Item	Qty	Notes
Heat pump + air handler	1 set	3.5 tons, cold-climate, variable speed, 5 kW strips
Supply registers	18	ceiling and high sidewall, with boots
Return grilles	3	filter grille in the hall, living and master hall
Transfer grilles / jump ducts	5	bedrooms and office
Trunk duct	114 ft	rigid sheet metal, sealed, R-8 wrap
Flex duct, 6-8"	236 ft	R-8, + 25 %
Return duct	~60 ft	14-16" R-8
ERV	1	133 CFM
Line set + pad + disconnect	1	~40 ft